



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Repository link (<https://github.com/Syed-56/SpaceX-falcon9-landing-prediction>)
- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

SUMMARY OF METHODOLOGIES

- Data Collection
- Data Wrangling
- Data Analysis and Visualization
- Model Training and Evaluation

SUMMARY OF ALL RESULTS

- Launch success improved over time, indicating increasing Falcon 9 reliability.
- Launch site influenced success rates, with some sites performing better than others.
- Payload mass showed a relationship with launch success, investigated through scatter plots.
- Booster reuse became increasingly common, demonstrating SpaceX's reusable-launch strategy.
- Geographical analysis showed the concentration and distribution of SpaceX launch sites.
- Machine-learning models were evaluated to determine how well launch success could be predicted from the available features.

Introduction

- SpaceX reuses the Falcon 9 first stage, cutting launch cost to ~\$62M vs \$165M+ for competitors
- Business question: can we predict landing success (and thus estimate launch cost) using public flight data?
- Framed as the capstone's "Space Y" scenario — a rival company assessing SpaceX's real cost structure

Section 1

Methodology

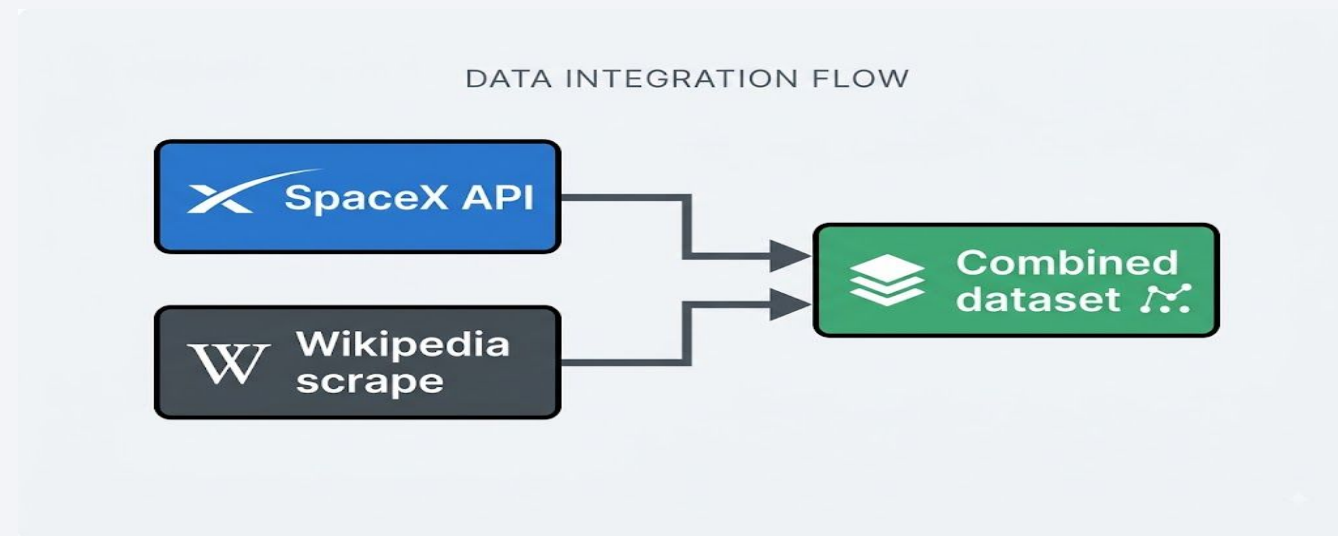
Methodology

Executive Summary

- Data collection methodology:
 - SpaceX REST API + Wikipedia web scraping
- Perform data wrangling
 - Cleaned nulls/duplicates, engineered binary Class target (1 = landed, 0 = did not land)
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models
 - Models Used : Logistic Regression, Random Forest, KNN, SVM

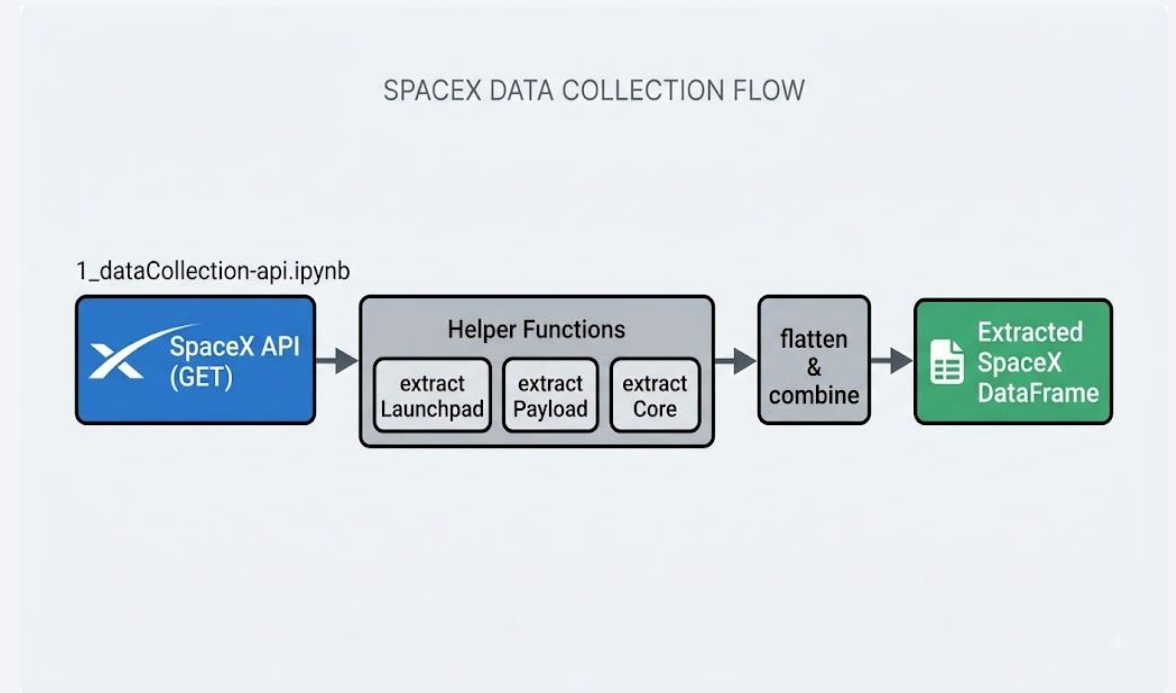
Data Collection

- Two sources: SpaceX REST API (structured launch/rocket/core data) + Wikipedia web scraping (historical Falcon 9 launch table via BeautifulSoup)
- Combined and cross-checked into a clean 90-row launch dataset



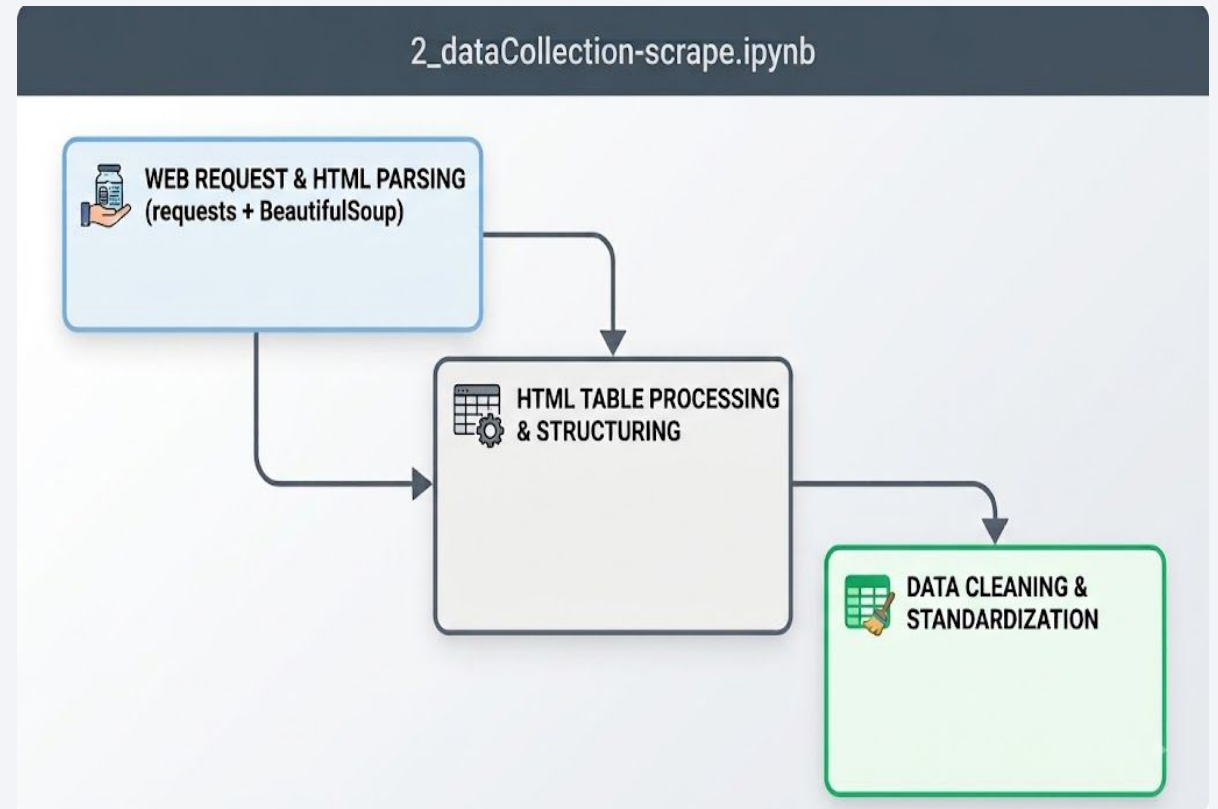
Data Collection – SpaceX API

- 1_dataCollection-api.ipynb: GET request to SpaceX API → extract rocket, launchpad, payload, and core data via helper functions) → flatten into a DataFrame
- [SpaceX-falcon9-landing-prediction/notebooks/1_dataCollection-api.ipynb at main · Syed-56/SpaceX-falcon9-landing-prediction](#)



Data Collection - Scraping

- 2_dataCollection-scrape.ipynb: pulled the Wikipedia "List of Falcon 9 and Falcon Heavy launches" table with requests + BeautifulSoup, parsed HTML table rows into structured columns, cleaned with unicodedata/regex
- [SpaceX-falcon9-landing-prediction/notebooks/2_dataCollection-scrape.ipynb at main · Syed-56/SpaceX-falcon9-landing-prediction](#)



Data Wrangling

- 3_dataWrangling.ipynb: handled missing values, standardized outcome labels
- Defined binary target: Class = 1 if landed successfully, 0 otherwise, derived from the Outcome column
- Result: 66.7% overall landing success rate
- [SpaceX-falcon9-landing-prediction/notebooks/3_dataWrangling.ipynb at main · Syed-56/SpaceX-falcon9-landing-prediction](#)

```
In [16]: print("Success Rate is ", df['Class'].mean()*100, "%")  
Success Rate is 66.66666666666666 %
```

EDA with Data Visualization

- Used seaborn/matplotlib to explore relationships between Flight Number, Payload Mass, Launch Site, Orbit, and landing outcome
- Chart types: numerical and categorical scatter plots (catplot), grouped bar chart (success rate by orbit), line chart (yearly trend)

[SpaceX-falcon9-landing-prediction/notebooks/5_dataViz.ipynb at main · Syed-56/SpaceX-falcon9-landing-prediction](#)

EDA with SQL

Loaded launch data into SQLite, ran 10 SQL queries covering: distinct launch sites, filtering by site prefix, payload aggregation by customer/booster, first successful landing date, boosters by payload range, outcome counts, and time-windowed outcome ranking

[SpaceX-falcon9-landing-prediction/notebooks/4_dataAnalysis-sql.ipynb at main · Syed-56/SpaceX-falcon9-landing-prediction](#)

Build an Interactive Map with Folium

Added launch site markers, circles marking site radius, and colored markers per launch outcome (green = success, red = failure) on a Folium map

Purpose: visualize whether launch sites cluster near coastlines/equator for geographic/safety reasons

[SpaceX-falcon9-landing-prediction/notebooks/6_mapsPlotting.ipynb at main · Syed-56/SpaceX-falcon9-landing-prediction](#)

Build a Dashboard with Plotly Dash

Built with spacex-dash-app.py: pie chart of launch success by site, dropdown to filter by site, range slider to filter by payload mass, scatter plot of payload vs. outcome

Purpose: let a non-technical stakeholder interactively explore which sites/payload ranges have the best success rates

[SpaceX-falcon9-landing-prediction/app/spacex-dash-app.py at main · Syed-56/SpaceX-falcon9-landing-prediction](#)

Predictive Analysis (Classification)

Standardized features with StandardScaler, 80/20 stratified train-test split

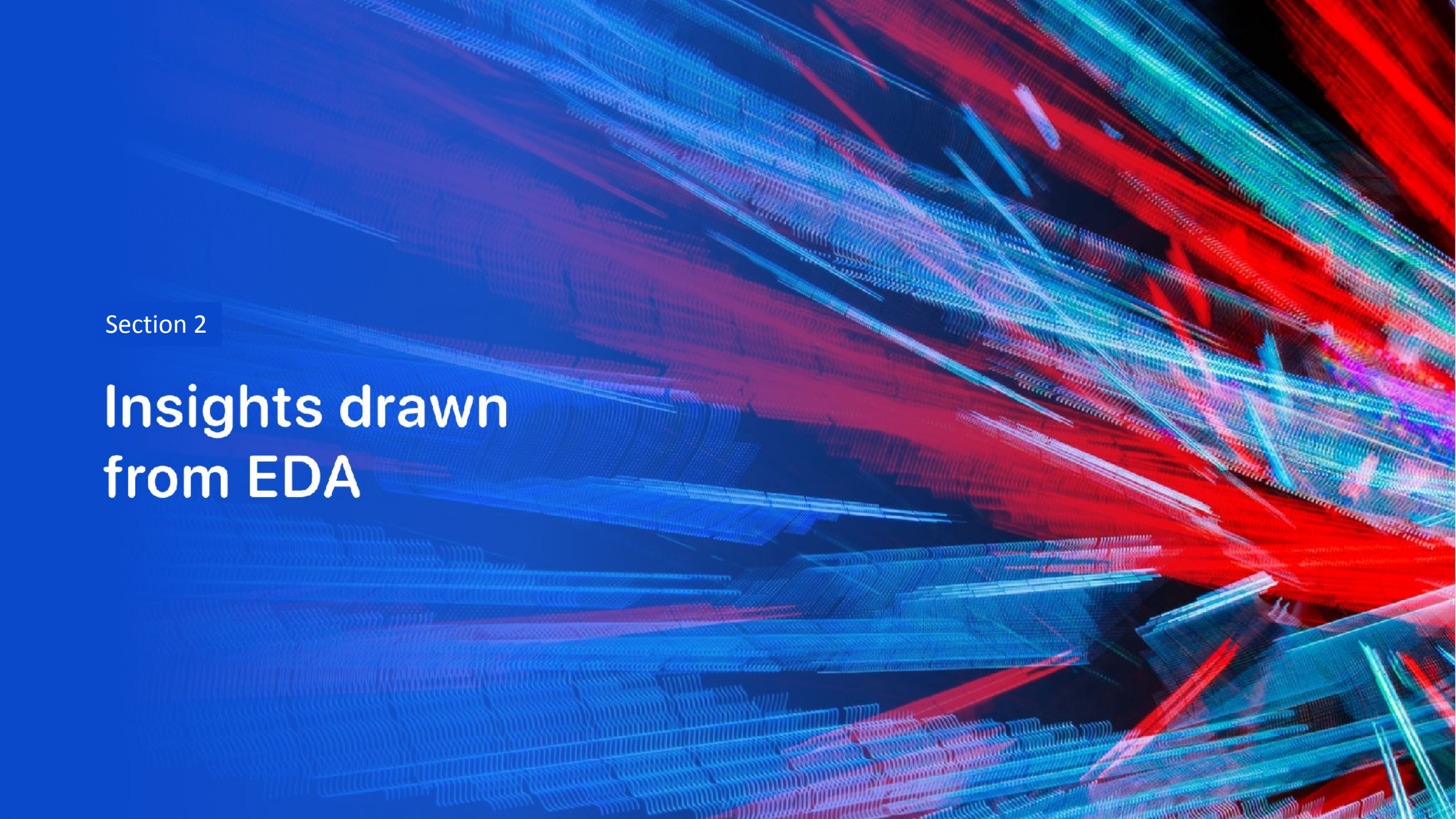
Trained 4 models — Logistic Regression, SVM, Decision Tree, KNN — each tuned via GridSearchCV (10-fold CV)

Selected best model by comparing cross-validation accuracy against held-out test accuracy (checking for overfitting)

[SpaceX-falcon9-landing-prediction/notebooks/7_modelDevelopment-evaluation.ipynb at main · Syed-56/SpaceX-falcon9-landing-prediction](#)

Results

- Exploratory Data Analysis: Analyzed 90+ Falcon 9 launches, examining launch success, payload mass, launch sites, and booster reuse. Results showed an overall improvement in launch reliability and highlighted differences in performance across launch sites.
- Interactive Analytics: Developed an interactive Dash dashboard with launch-site and payload-range filters, allowing users to explore success/failure outcomes, payload distributions, and launch-site performance dynamically.
- Predictive Analysis: Built and compared multiple classification models to predict launch success using launch characteristics. Model performance was evaluated using accuracy and confusion matrices to identify the best-performing model.

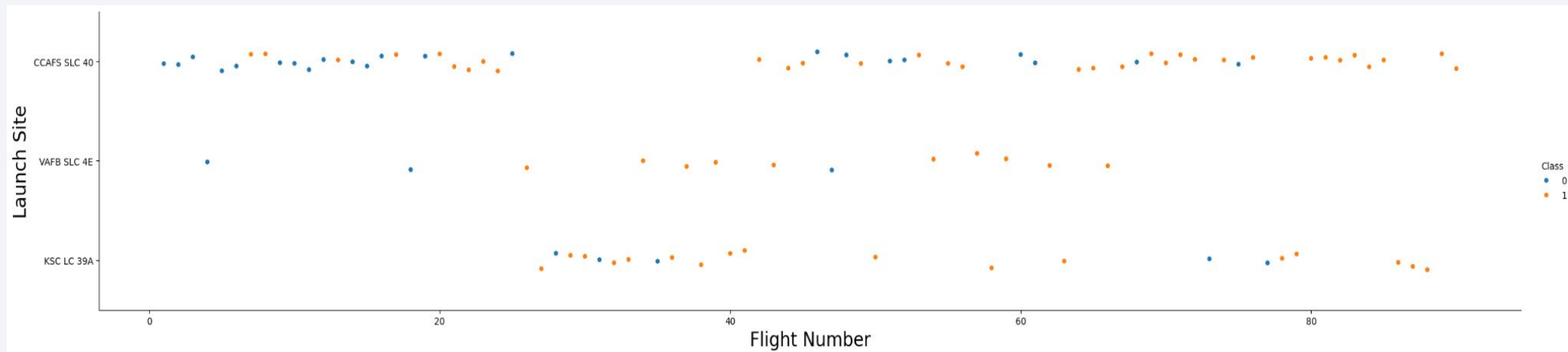
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue and red on the right. These streaks are layered over a fine, light-colored grid or mesh pattern, creating a sense of depth and digital complexity.

Section 2

Insights drawn from EDA

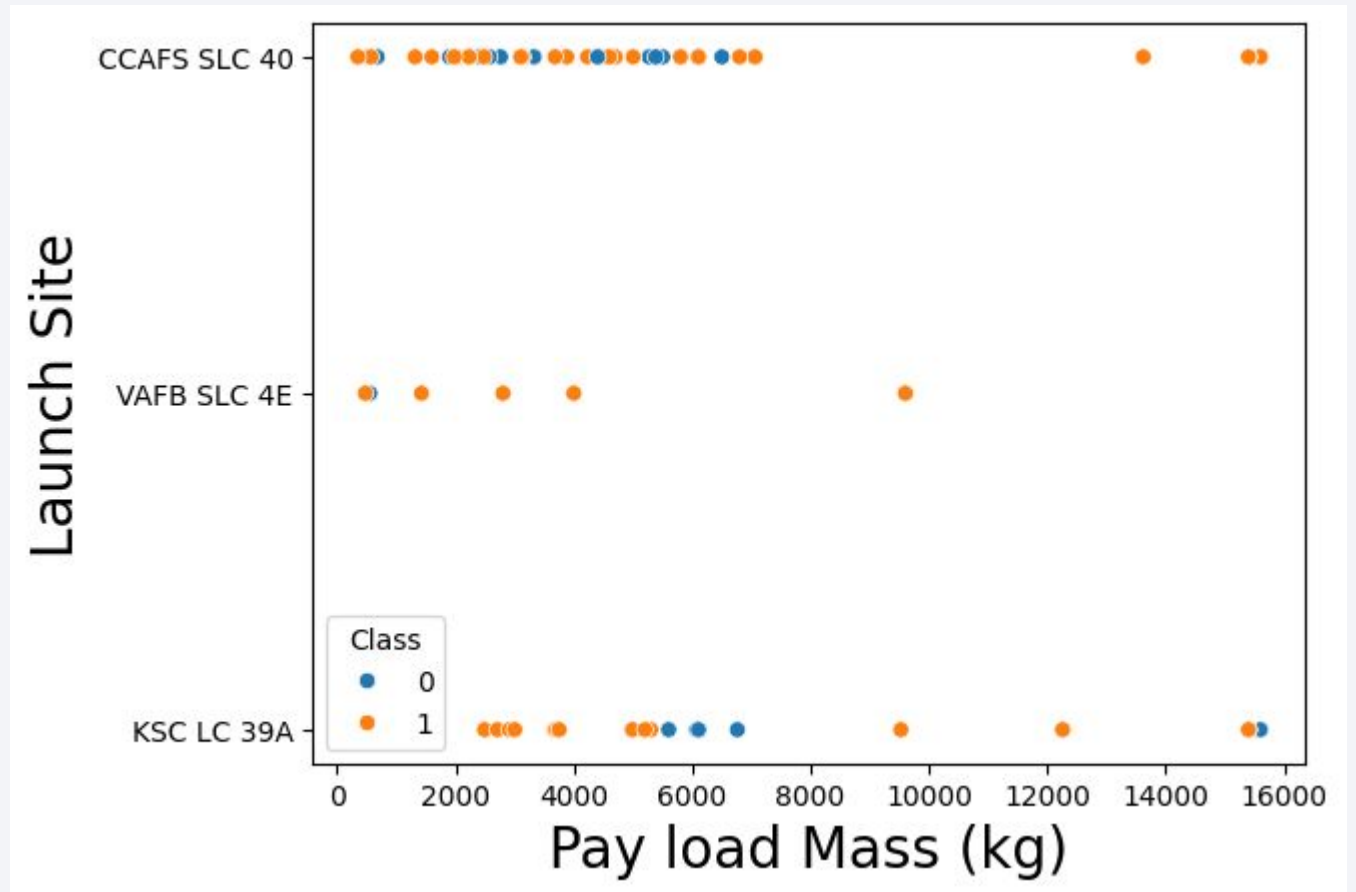
Flight Number vs. Launch Site

- flight number don't show a clean single-launch-site pattern, but you can see which sites handled which mission types over time.



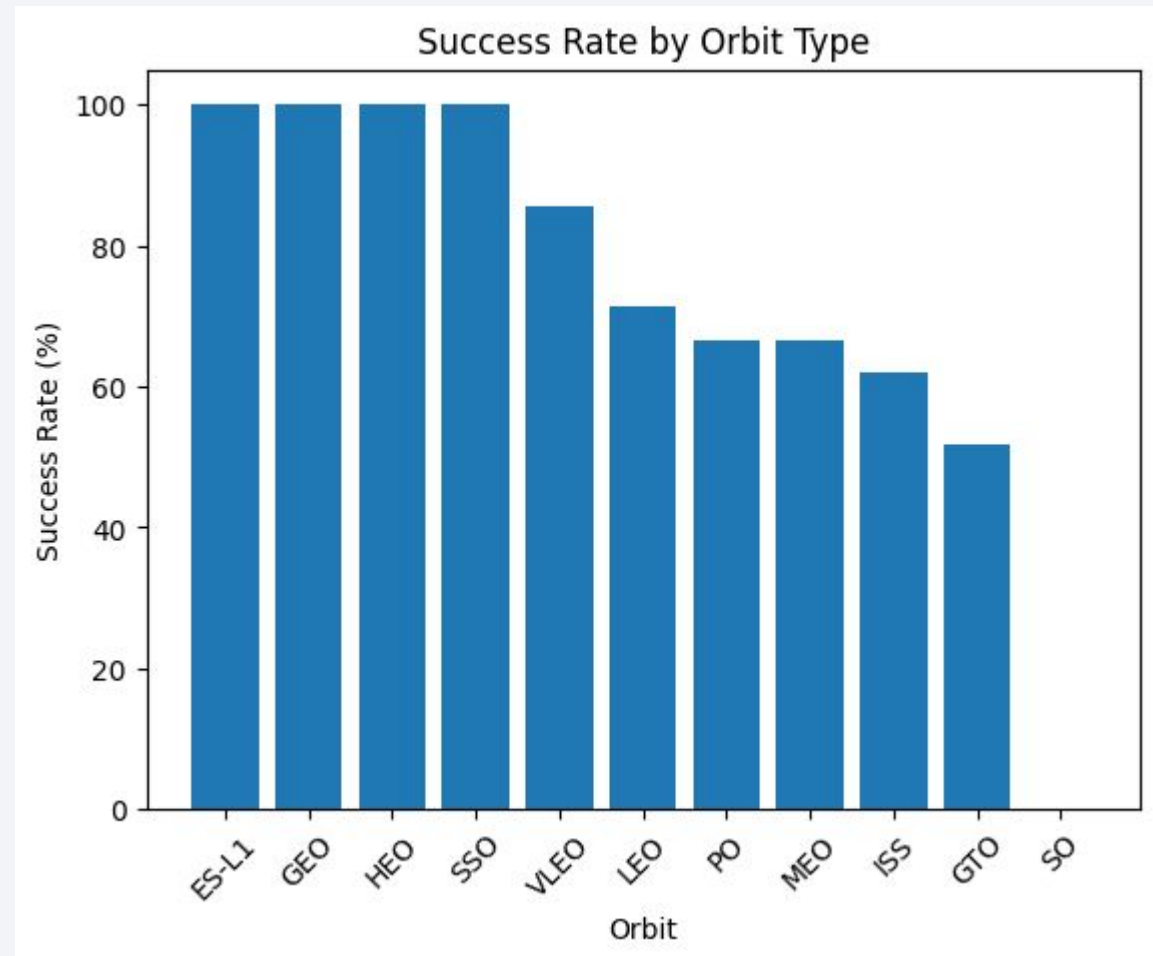
Payload vs. Launch Site

- Heavier payloads cluster at specific sites (VAFB tends to handle lighter/polar-orbit payloads vs. CCAFS/KSC).



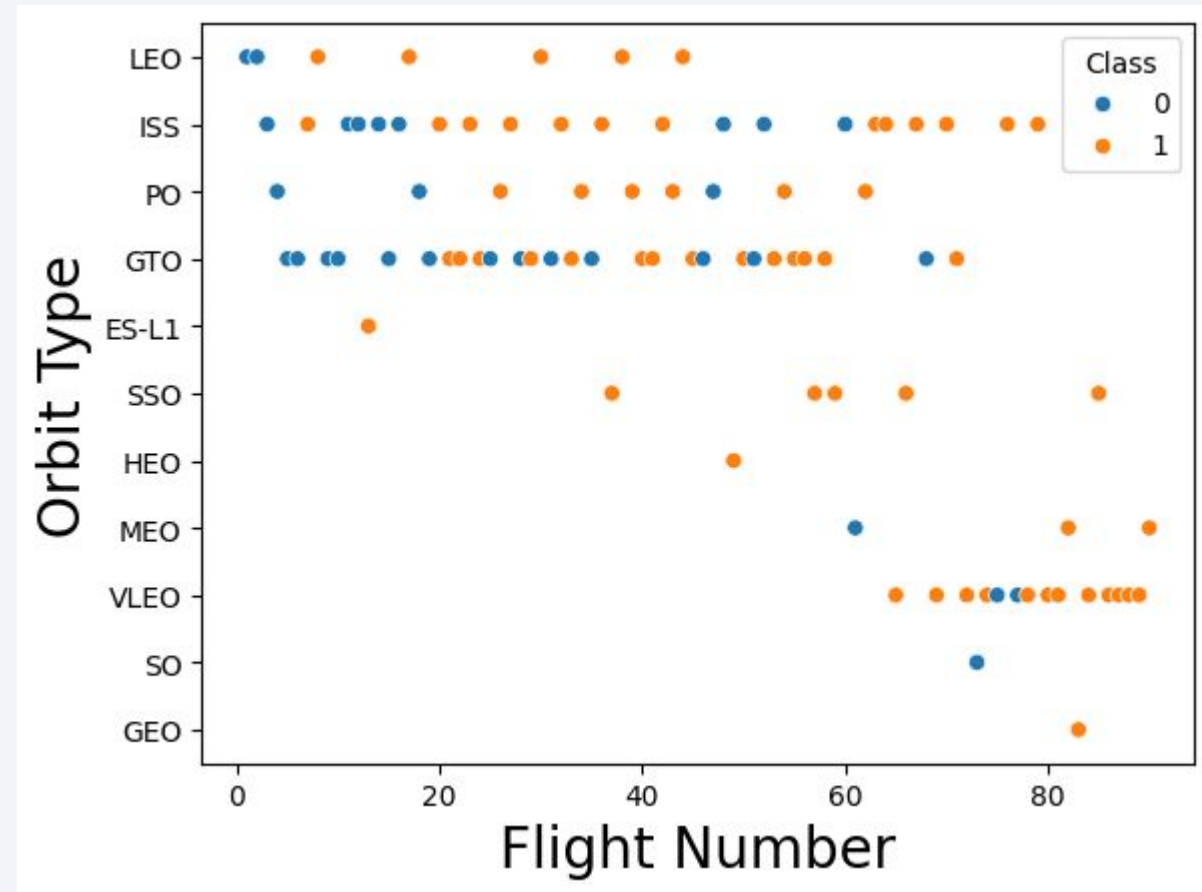
Success Rate vs. Orbit Type

- Success rate varies sharply by orbit
- ES-L1/GEO/SSO/HEO near 100% success rate while others are slightly below



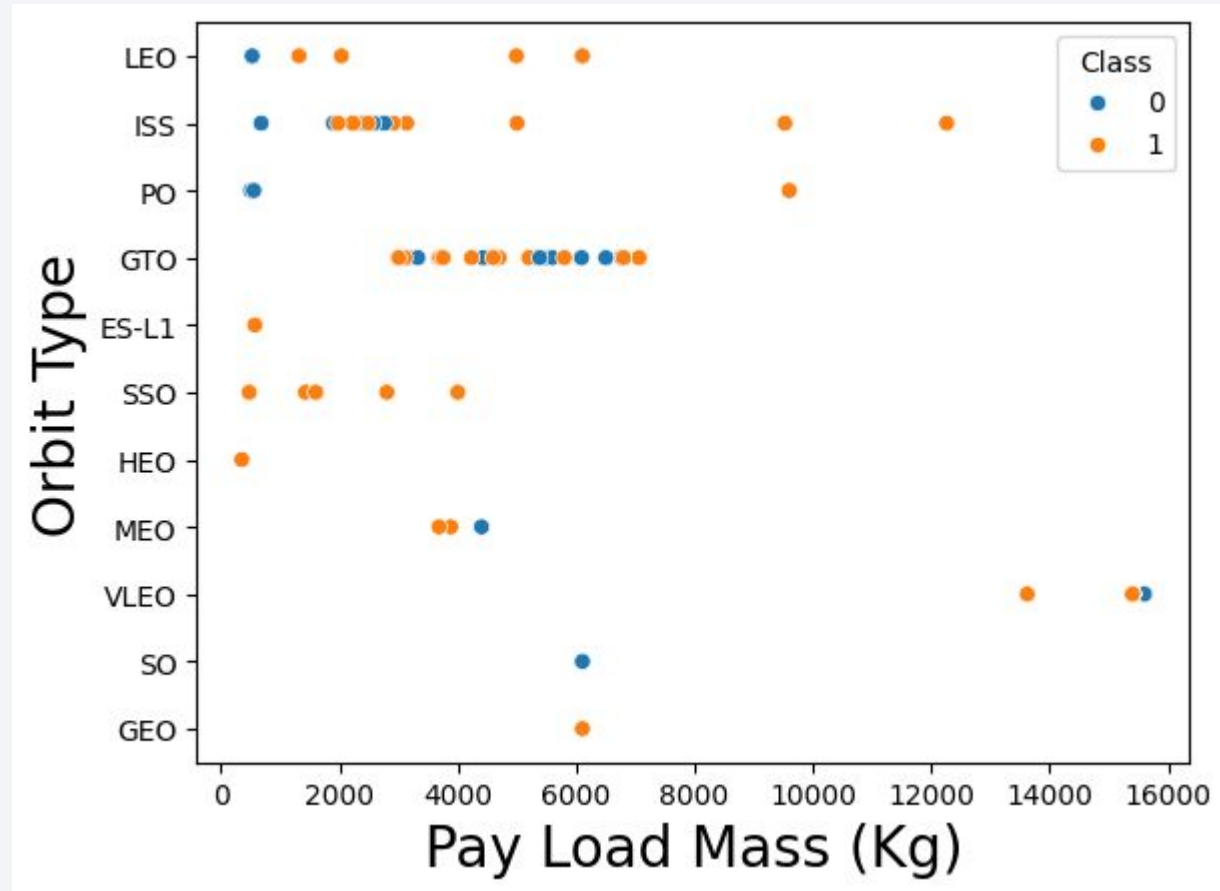
Flight Number vs. Orbit Type

- Later flight numbers show more successful landings across most orbit types which is an evidence of SpaceX's improving technology over time.



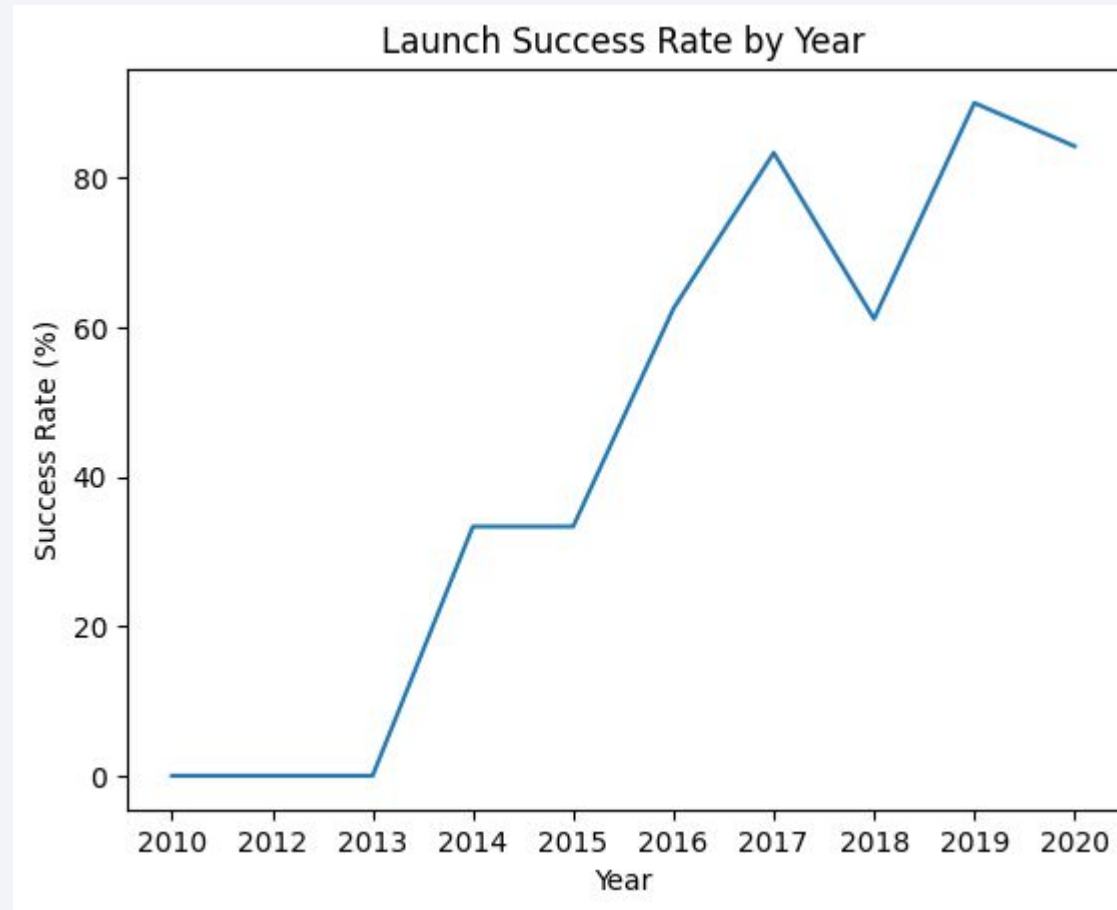
Payload vs. Orbit Type

- Heavier payloads reduce landing success likelihood, especially for GTO orbit; lighter LEO/ISS payloads land more reliably.



Launch Success Yearly Trend

- Success rate trends upward year over year as SpaceX iterated on landing technology.



The background of the slide is an abstract composition of numerous thin, overlapping lines and streaks in shades of blue, red, and teal. These lines are oriented diagonally, creating a sense of motion and depth. The overall effect is reminiscent of a digital data stream or a high-speed optical trail.

Section 3

Data Analysis with SQL

All Launch Site Names

- 4 distinct sites: CCAFS LC-40, CCAFS SLC-40, VAFB SLC-4E, KSC LC-39A

```
: %sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
* sqlite:///my_data1.db
Done.
: Launch_Site
-----
  CCAFS LC-40
  VAFB SLC-4E
  KSC LC-39A
  CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

- First 5 records from CCAFS sites — shows early mission details (Dragon Spacecraft Qualification, demo flights)

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

* sqlite:///my_data1.db
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total NASA (CRS) payload mass = 45,596 kg

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACE_TABLE WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db  
Done.
```

```
Total_Payload_Mass
```

```
45596
```

Average Payload Mass by F9 v1.1

- Average payload mass for booster F9 v1.1 = 2,928.4 kg

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) AS Avg_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

<u>Avg_Payload_Mass</u>

2928.4

First Successful Ground Landing Date

- 22 December 2015

```
%sql SELECT MIN(Date) AS First_Successful_Ground_Landing FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
First_Successful_Ground_Landing
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- Boosters: F9 FT B1022, B1026, B1021.2, B1031.2

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Success: 98, Success (payload status unclear): 1, Failure (in flight): 1

```
%sql SELECT Mission_Outcome, COUNT(*) AS Total FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- 12 boosters tied at max payload
- (all F9 B5 variants — B1048.x, B1049.x, B1051.x, B1056.4, B1058.3, B1060.x)

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- 2 drone-ship failures in 2015: Jan (F9 v1.1 B1012, CCAFS LC-40), Apr (F9 v1.1 B1015, CCAFS LC-40)

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- No attempt (10) > Success drone ship (5) = Failure drone ship (5) > Success ground pad (3) = Controlled ocean (3) > Uncontrolled ocean (2) = Failure parachute (2) > Precluded drone ship (1)

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is dark blue with a thin white line representing the horizon. The city lights are visible as bright yellow and orange spots against the dark blue background of the night sky.

Section 3

Launch Sites Proximities Analysis

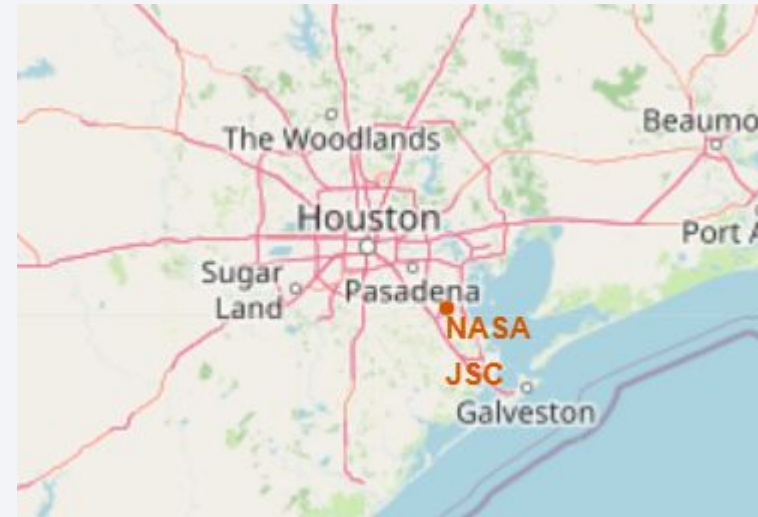
ALL Launch SITES in US

- VAFB SLC-4E was launched from California
- All other ones were launched from Florida



NASA JSC HQ

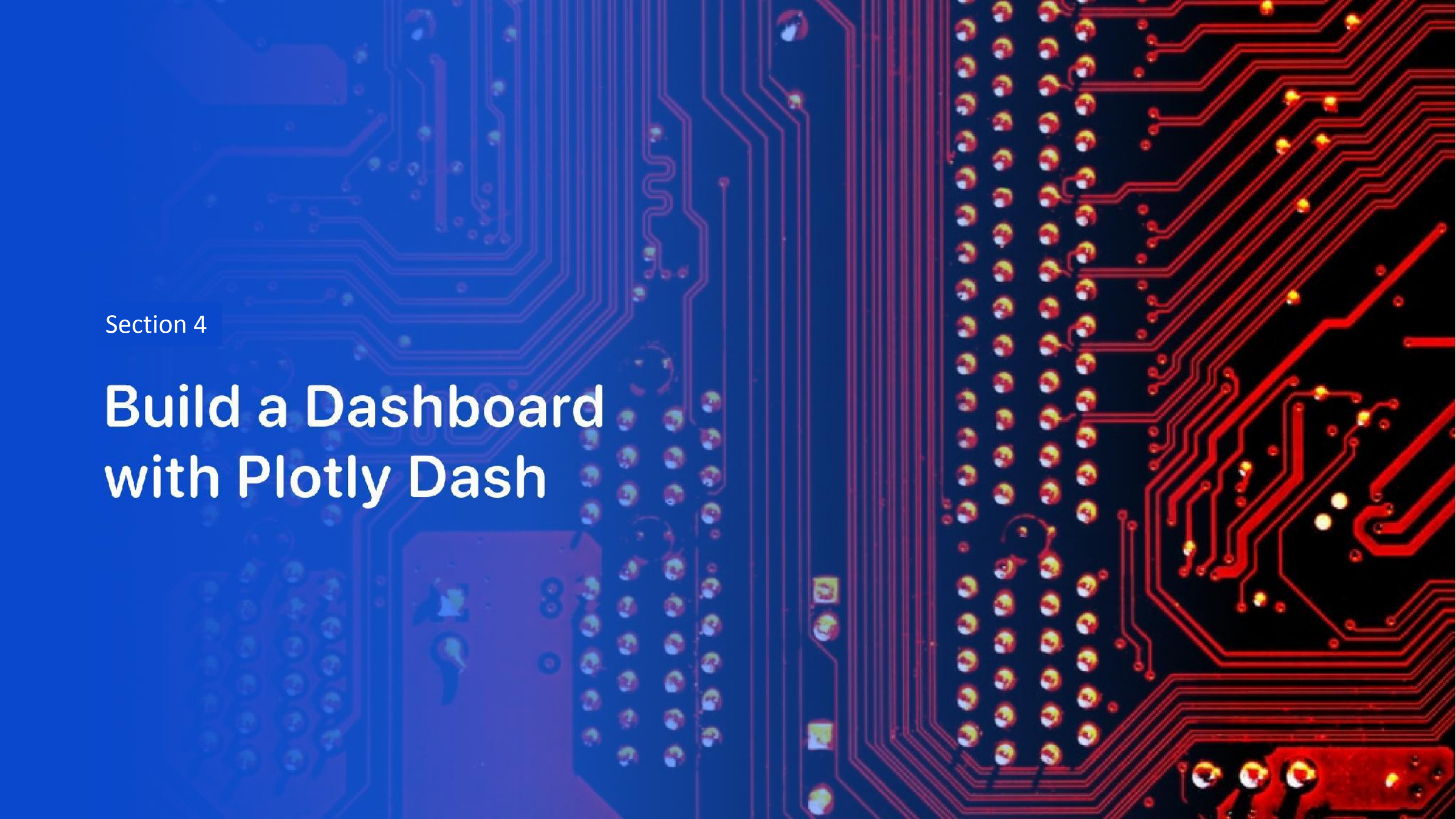
Initial center location to be NASA
Johnson Space Center at Houston, Texas.



Total Amounts of Launch from each site

California has only 10 launch (since only 1 location) but Florida has 46 launches across 3 sites





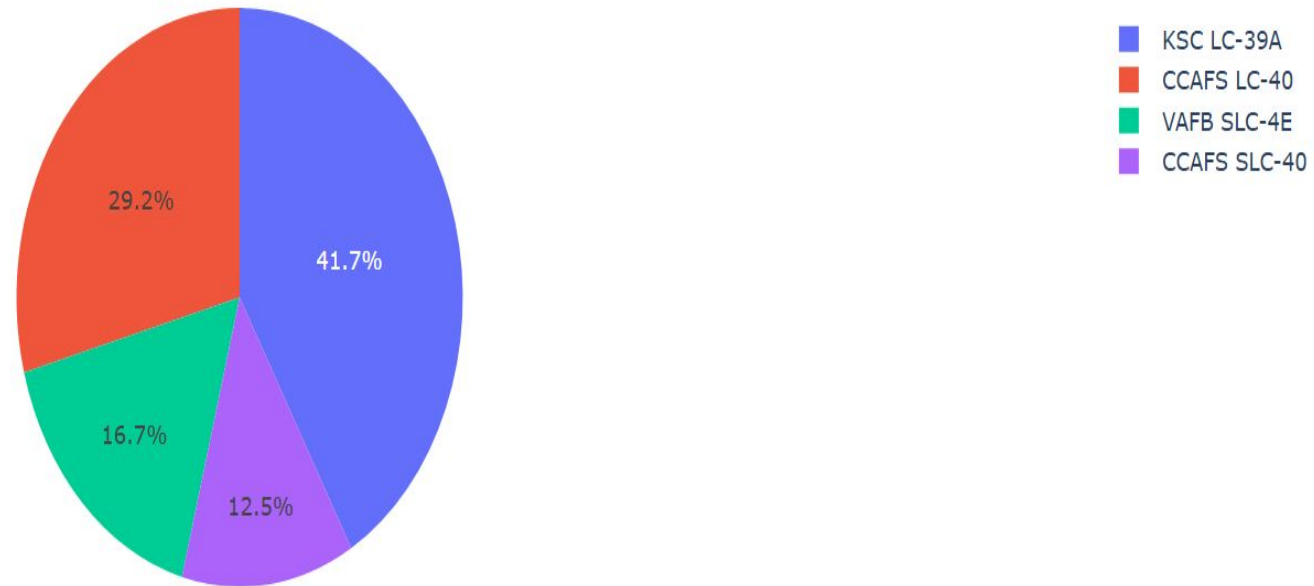
Section 4

Build a Dashboard with Plotly Dash

Contribution of Each Launch Site in Successful Launches

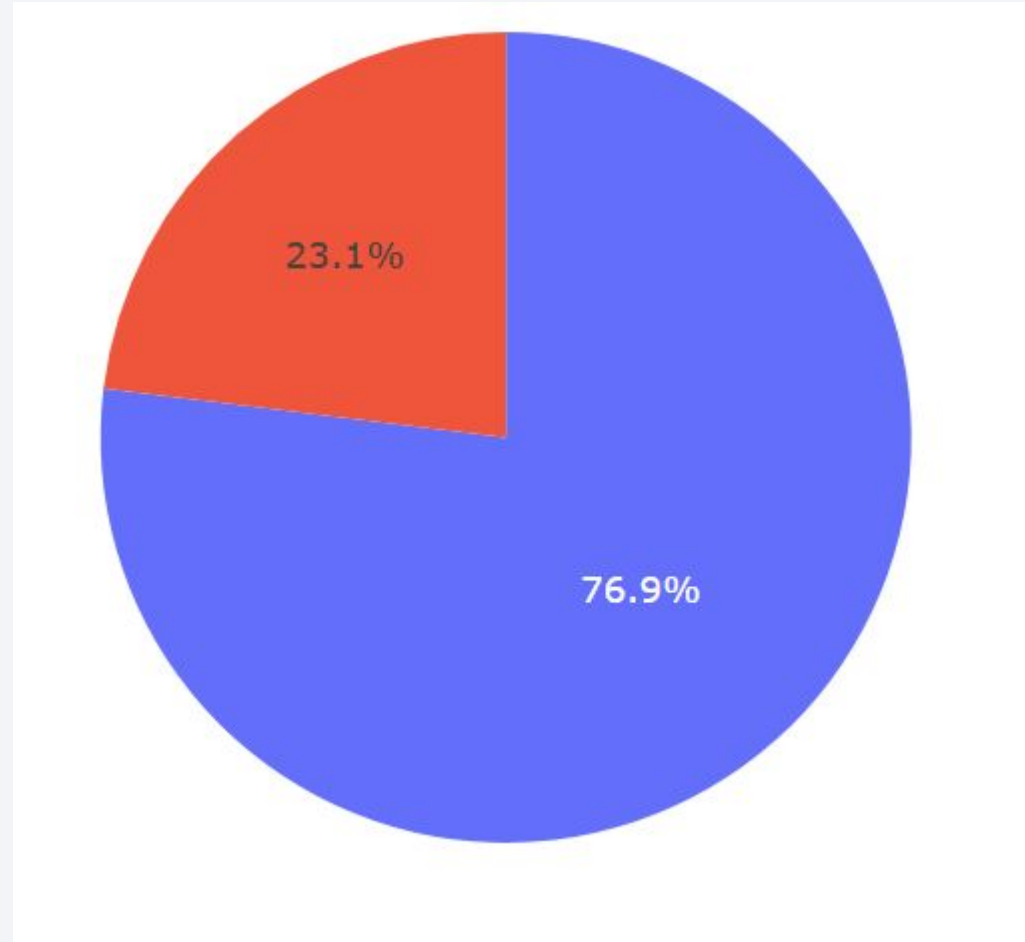
KSC LC-39A contribute to the most successful launches

Total Successful Launches by Site



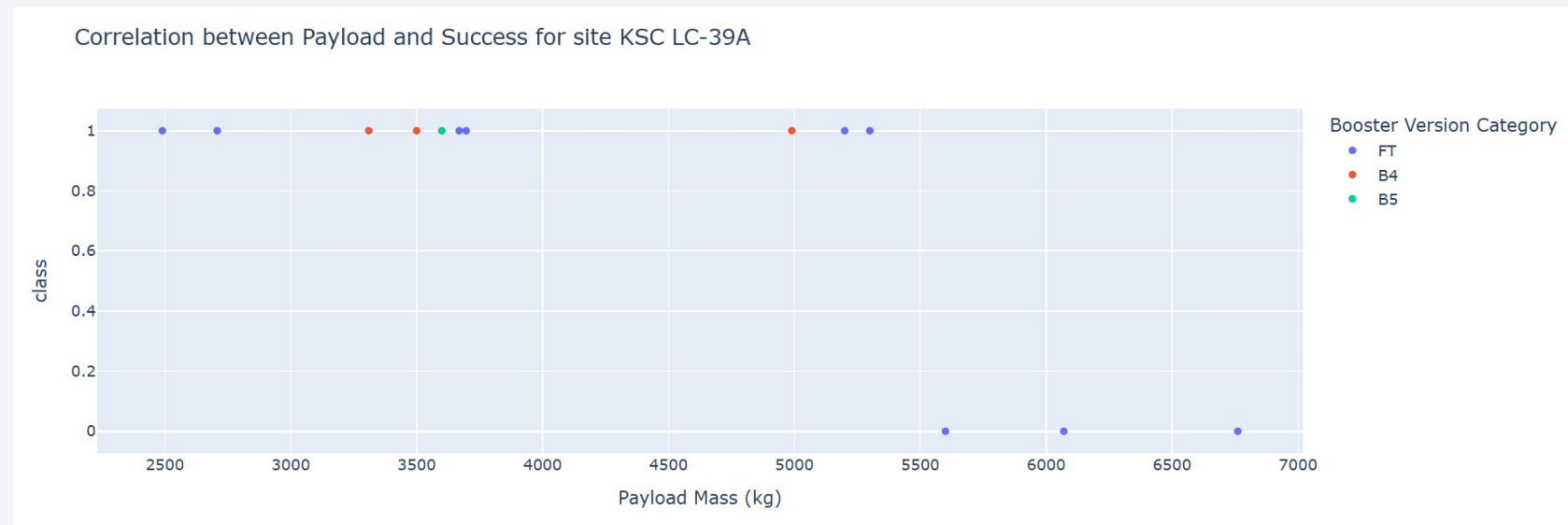
Success Comparison of KSC LC 39-A Launches

- Success vs Failure Ratio of KSC LC 39-a Launch Site.
- Almost 77% launches are successfull



Correlation between Payload and Success Rate at KSC LC-39A

- Payloads below ~5,000 kg were mostly associated with successful launches, while failures were concentrated at higher payload masses.
- At KSC LC-39A, the plot suggests a negative relationship between payload mass and launch success, although the sample size is limited.



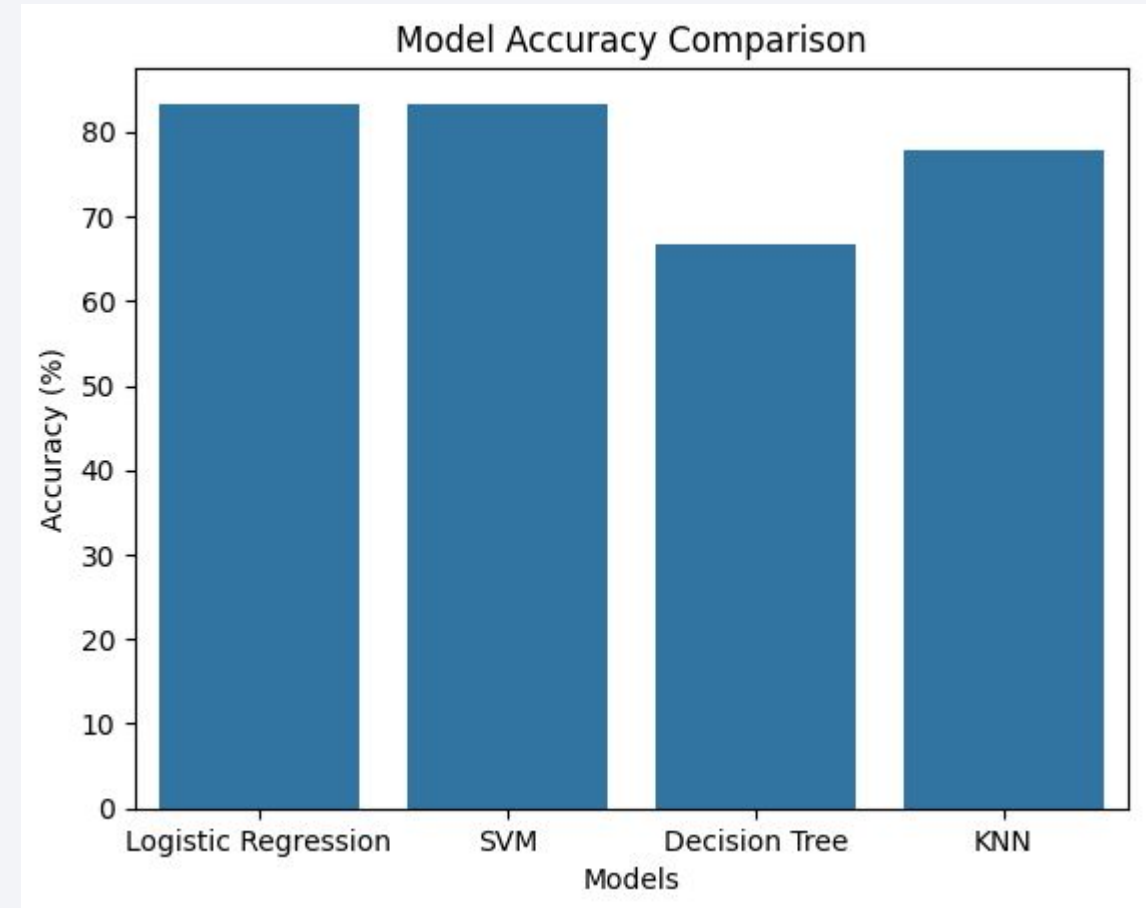


Section 5

Predictive Analysis (Classification)

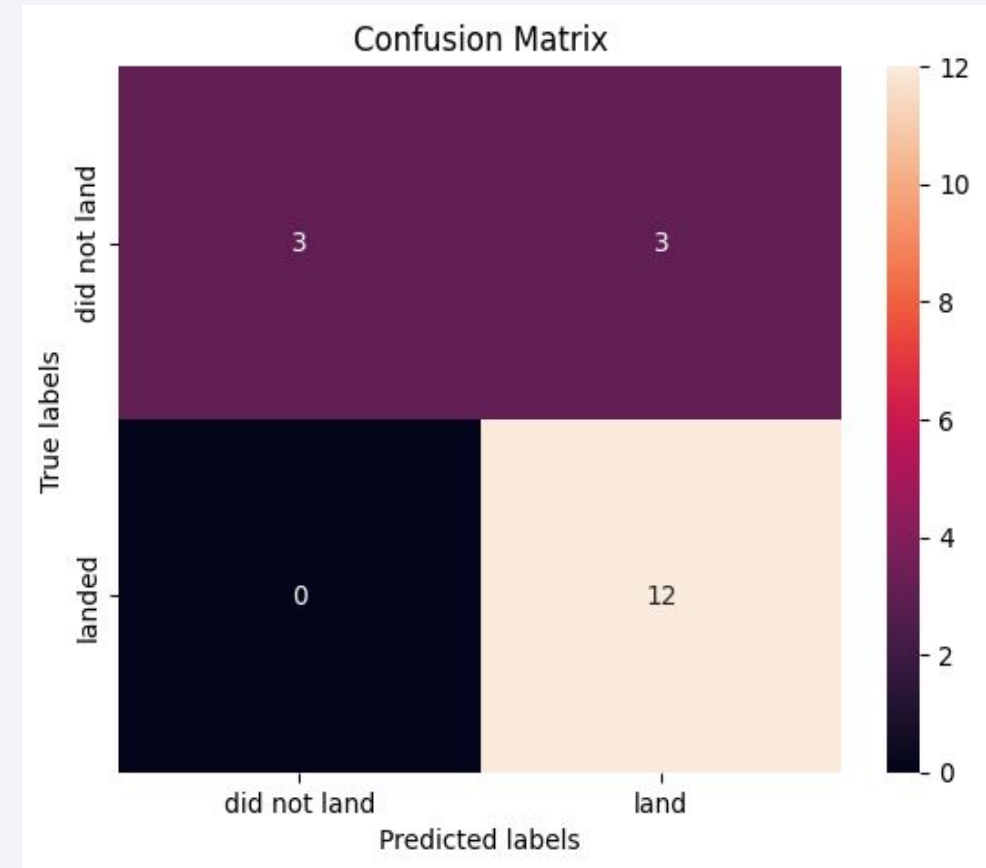
Classification Accuracy

- Logistic Regression and SVM tied for the highest test accuracy at 83.3%, demonstrating the best performance on the test dataset.
- Decision Tree yielded the lowest accuracy at approximately 66.7%, while KNN achieved intermediate performance at 77.8%.



Confusion Matrix

- The model achieved an overall accuracy of 83.3% (15 out of 18 correct predictions), successfully predicting all 12 actual landing events with zero false negatives.
- It shows a slight bias toward predicting successful landings, incorrectly misclassifying 3 out of 6 "did not land" instances as "land" (false positives).



Conclusions

- Falcon 9 landing success can be predicted with ~83% accuracy using only public flight data
- Payload mass and orbit type are the strongest predictors of landing outcome
- Landing success has improved significantly over time as SpaceX matured its recovery process
- Logistic Regression and SVM generalize best; Decision Tree overfits on this small (90-row) dataset
- Limitation: small sample size — more data or additional features (weather, sea state) could improve robustness

Appendix

Logistic Regression Hyperparameters

GridSearchCV ⓘ ?		
▼ Parameters		
estimator	LogisticRegression()	
param_grid	{ <code>'C': [0.01, 0.1, ...]</code> , <code>'penalty': ['l2']</code> , <code>'solver': ['lbfgs']</code> }	
scoring	None	
n_jobs	None	
refit	True	
cv	10	
verbose	0	
pre_dispatch	'2*n_jobs'	
error_score	nan	

LogisticRegression(C=0.1, penalty='l2') ?		
▼ LogisticRegression ?		
▼ Parameters		
penalty	'l2'	
C	0.1	
l1_ratio	0.0	
dual	False	
tol	0.0001	
fit_intercept	True	
intercept_scaling	1	
class_weight	None	

Thank you!

